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Ask



Getting started with Nvidia Jetson Nano SD-CARD

In this guide, we will help you get started with balenaCloud by:

- Setting up your **Nvidia Jetson Nano SD-CARD** device and bringing it online on the balenaCloud dashboard.
- Deploying a *hello world* project on the device in your language of choice.
- Developing the sample project: making changes and testing them on the device in real-time.

Once you've completed this getting started guide to balena, you'll be equipped with the fundamentals needed to continue developing your application using balenaCloud and be on the path to deploying fleets of devices to production. If you are looking for definitions of certain terms, refer to the [glossary](#).

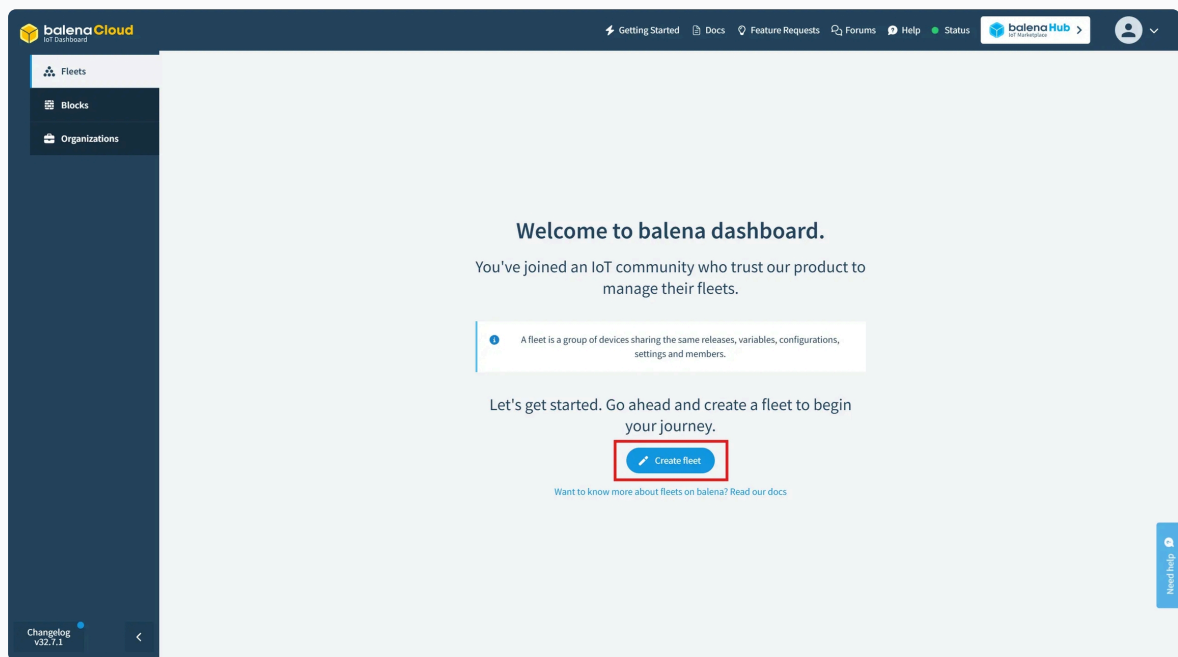
What you'll need

- A [Nvidia Jetson Nano developer kit](#) ↗
- A 16GB or larger microSD card. The [speed class](#) ↗ of the card also matters - UHS speed class I or above is recommended.
- An ethernet cable (the Jetson Nano developer kit does not include WiFi but it can be added via an optional mPCIe card or USB dongle).
- Compatible power supply - a 5V 4A barrel jack PSU is recommended, but you can also use a 5V 2.5A micro USB supply.
- A [balenaCloud account](#) ↗.

Create a fleet

A fleet is a group of devices that share the same [architecture](#) and run the same code. Devices are added to fleets and can be moved between fleets at any time.

To create your first fleet, log into your [balenaCloud dashboard](#) and click the **Create fleet** button.



Enter a fleet name, select the **Nvidia Jetson Nano SD-CARD** device type, and click **Create new fleet**:

Create fleet

Organization

unicorn's Organization

Fleet

First-Fleet

A good fleet name is concise and easy to recall. Looking for ideas? What about **wicked-glitter**?

Default device type

Choose a device type...

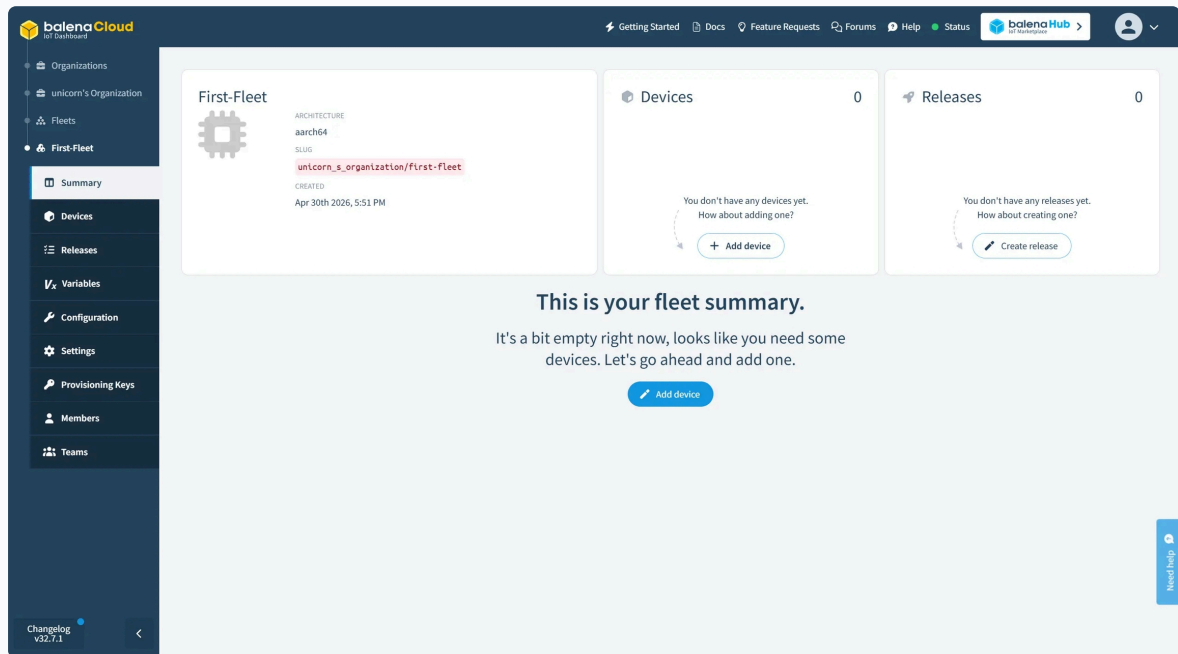
☐ Show discontinued device types

[I don't have a device on hand](#)

Create new fleet

You'll then be redirected to the summary of the newly created fleet, where you can add your first Nvidia Jetson Nano SD-CARD.

Add a device and download OS



balenaCloud builds a custom balenaOS image configured for Nvidia Jetson Nano SD-CARD which allows the device to provision and join the new fleet you created automatically. Start by clicking **Add device** on the fleet summary. Your device type will be preselected here since you already chose it when creating the fleet. Other device types of the same [architecture](#) can also be picked to join the fleet.

Add new device

Device type 

 Choose a device type... 

OS version

4.0.16  RECOMMENDED ☐ Show outdated versions

Edition

☐ Development  Recommended for first time users

Development images should be used when you are developing an application and want to use the fast [local mode](#) workflow.

☒ Production

Production images are ready for production deployments, but don't offer easy access for local development.

Select an OS type of *balenaOS*, and you will see a list of available balenaOS versions with the latest preselected. Choose a **Development** version of the OS. The production OS does not facilitate the development workflow we'll be using. Find out more about the [differences between Development and Production images](#).

Network

☐ Ethernet only

☒ Wifi + Ethernet

WiFi SSID

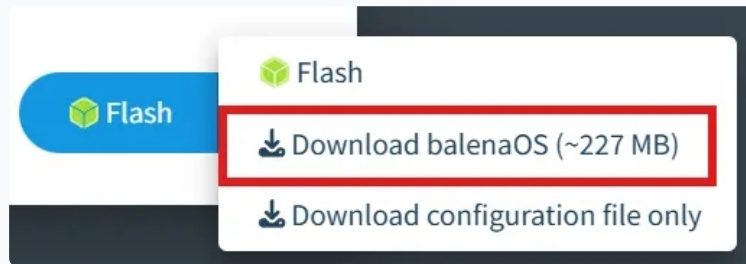
Wifi Passphrase



Select the type of network connection you'll be using: *Ethernet Only* or *Wifi + Ethernet*. A network connection is required to allow the device to connect to balenaCloud. Selecting *Wifi + Ethernet* allows you to enter a *Wifi SSID* and *Wifi Passphrase* which is then built into the image.



Click the arrow by the **Flash** button. You'll see several options.



Pick the option that works best for you:

Flash

Download balenaOS

Download configuration file only

If you have [Etcher](#) installed, you can just click the Flash button. This will automatically open Etcher with your image loaded. The contents of the image that gets loaded are the same as the contents of the image you would get from the **Download balenaOS** option. Once Etcher opens, proceed to [Provision device](#).

This option is best for quickly flashing a device(s) in a single session. Once you close Etcher, the image will no longer be loaded in it.

Once you have chosen one of the options, the button should update to whichever option you have chosen. Click the button and proceed.

Provision device

Next, we will flash the downloaded image onto the device. To do so, follow the following steps:

- To provision Nvidia Jetson Nano SD-CARD, follow the instructions using our [Jetson Flash tool](#) to make the process more streamlined.

When complete, after a minute or two the device should appear on your balenaCloud [dashboard](#), and you should now be ready to deploy some code. If you are not able get the device to appear on the dashboard, then check out our [troubleshooting guide for Nvidia Jetson Nano SD-CARD](#) or try our [support channels](#).

Install the balena CLI

Now that you have an `operational` device in your fleet, it's time to deploy some code. We will use the balena CLI for this. Follow the instructions below to install balenaCLI for the operating system available on your system. Skip the next part if you have balena CLI already installed.

OSX

Windows

Linux

1. [Download the CLI installer](#).
2. Double click the downloaded file to run the installer and follow the installer's instructions.
3. To run balena CLI commands, open the Terminal app ([Terminal User guide](#)).

After balena CLI is installed, login to your balena account using the `balena login` command on the terminal:

```
$ balena login
```

```

| | _ _ _ _ _ | | _ _ _ _ _ | | _ _ _ _ _
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```

Logging in to cloud.com

? How would you like to login? (Use arrow keys)

- › Web authorization (recommended)

Credentials

Authentication token

I don't have a balena account!

You will be asked to choose an authentication method, choose *Web authorization* which will bring up a web browser window that allows you to login to your balenaCloud account. Click the **Authorize** button, and head back to the terminal after the login successful message appears.



Balena CLI

By clicking authorize you share your authentication token with the balena CLI in a secure way.

Authorize

Create a release

After login, test the balena CLI by running the `balena fleet list` command, which should return information about the fleet you created in the previous step. Take a note of the fleet `NAME` as you'll need this in the next step to push the code to your device(s) in that fleet.

```
$ balena fleets
ID      NAME      SLUG      DEVICE
TYPE      DEVICE COUNT  ONLINE DEVICES
98264 First-Fleet  balena CLI/first-fleet  Nvidia Jetson Nano
SD-CARD    0            0
```

Node.js

Python

C++

Rust

Go

A nice project to try is the [balena-nodejs-hello-world ↗](#) project. It's a Node.js web server that serves a static page on port 80. To get started, [download the project ↗](#) as a zipped file from GitHub, unzip it and open a terminal in the root of the extracted project directory.

To create a release, use the `balena push First-Fleet` command replacing `First-Fleet` with the name of your fleet. Ensure you are working from the root of the extracted project directory.

```
$ balena push First-Fleet
```

This command pushes the code to the balena builders, where it will be compiled, built, turned into a release, and applied to every device in the fleet.

You'll know your code has been successfully compiled and built when our friendly unicorn mascot appears in your terminal:

Service	Image Size	Build Time
hello-world	190.04 MB	50 seconds

Build finished in 1 minutes, 4 seconds

```

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```

9

The screenshot displays the Balena Cloud dashboard for a device named 'new-wave'. The left sidebar shows the device's status as 'Updating' and 'Downloading'. The main panel displays various system metrics and configuration options. The 'Public Device URL' toggle is currently turned off. The 'Logs' section shows the device's boot process, including the release of update locks and the download of the 'balena-hello-world' service. The 'Terminal' section shows a 'Start terminal session' button.

Device Details:

- NAME:** new-wave
- STATUS:** Updating
- HEARTBEAT:** Green
- UPDATE STATUS:** Downloading
- OS VARIANT:** DEVELOPMENT
- TARGET RELEASE:** 2.0.5
- LOCAL MODE:** Off
- LOCAL IP ADDRESS:** 192.168.86.97
- PUBLIC IP ADDRESS:** 94.174.218.77
- MAC ADDRESS:** E4:5F:01:E3:91:53
- CLOUDLINK:** Connected (for 3 minutes)
- HOST OS VERSION:** balenaOS 5.3.7
- CURRENT RELEASE:** 2.0.5+rev2
- IS ACTIVE:** On

Logs:

```
* Debug mode: off
* Running on all addresses.
WARNING: This is a development server. Do not use it in a production deployment.
* Running on http://192.17.0.2:80/ (Press CTRL+C to quit)
Releasing update locks
Supervisor starting
Supervisor starting
Supervisor starting
Creating network 'default'
Downloading image 'registry2.balena-staging.com/v2/dd352cc9dd9e4581c19f7b54a0220eddbsha256:ac22e6bbd80828a3a292426bf79fa6517eeabf0e6666a6b601bddd321462507'
Taking update locks
```

Terminal:

Select a target

Start terminal session

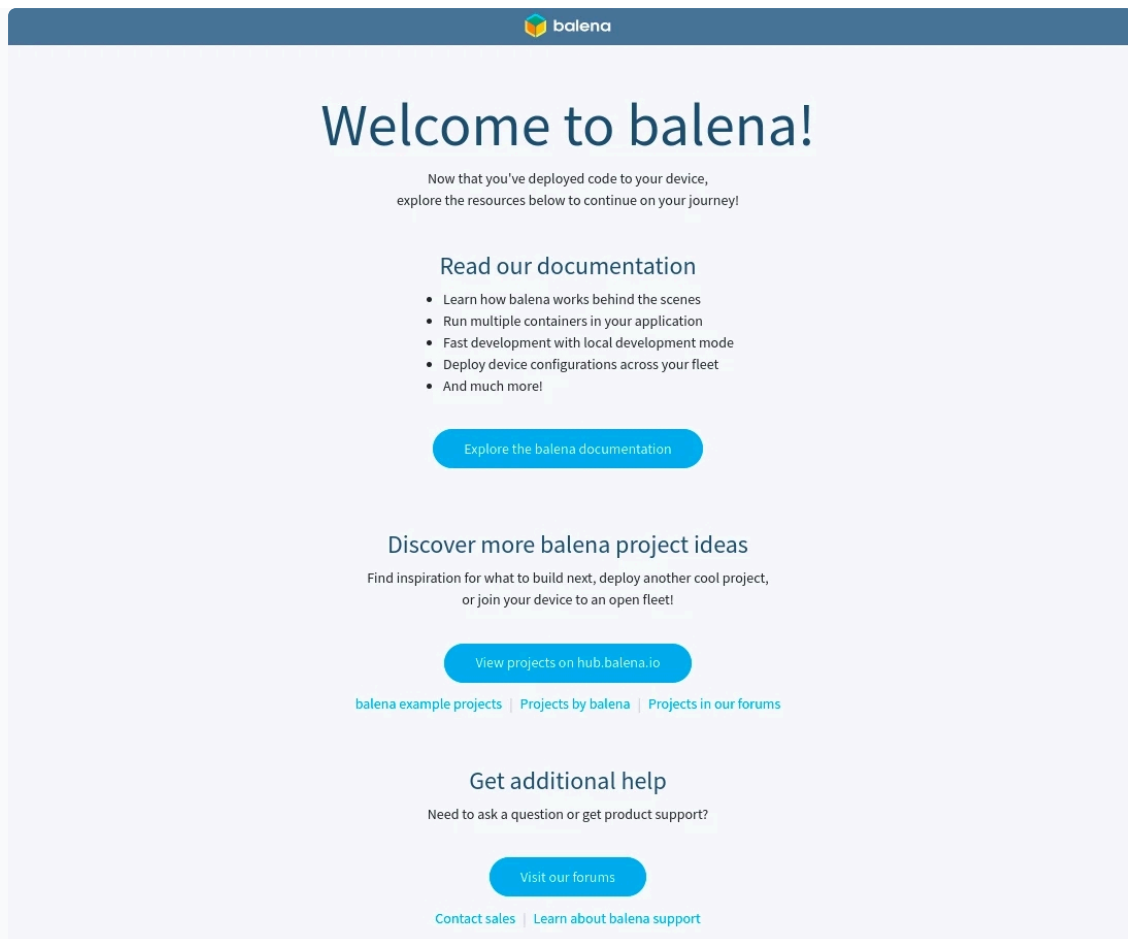
After the download, you should now have a web server running on your device and see some logs on your dashboard.

To give your device a public URL, click the *Public Device URL* toggle on the device dashboard. Public device URL allow you to serve content from the device to the world easily without configuration as long as the server is running on port 80.

The screenshot shows the Balena Cloud interface for a device named 'new-wave'. The device is in the 'FLEET' and is associated with the user 'shaun_mulligan/testing'. The status is 'Operational' with a green checkmark. The heartbeat is also green. The update status is 'Done'. The OS variant is 'DEVELOPMENT'. The target release is '2.0.5'. The local mode is set to 'Local' with a star icon. The local IP address is '192.168.86.97' and the public IP address is '94.174.218.77'. The MAC address is 'E4:5F:01:E3:91:53'. The supervisor version is '16.7.1' and the current release is '2.0.5'. The support access is 'Off'. The public device URL is highlighted with a red box and a toggle switch. The device is connected to the cloud link for 10 minutes. There are no tags configured yet and no notes are present.

Property	Value
FLEET	shaun_mulligan/testing
STATUS	Operational
HEARTBEAT	✓
UPDATE STATUS	Done
OS VARIANT	DEVELOPMENT
TARGET RELEASE	2.0.5
LOCAL MODE	Local
LOCAL IP ADDRESS	192.168.86.97
PUBLIC IP ADDRESS	94.174.218.77
MAC ADDRESS	E4:5F:01:E3:91:53
CLOUDLINK	✓
CLOUDLINK LAST CONNECTED	Connected (for 10 minutes)
UUID	0ac39cf
UPDATE STATUS DURATION	6 minutes
HOST OS VERSION	balenaOS 5.3.7
SUPERVISOR VERSION	16.7.1
CURRENT RELEASE	2.0.5 ✓
SUPPORT ACCESS	Off
IS ACTIVE	✓
PUBLIC DEVICE URL	☐
TAGS (0)	No tags configured yet
NOTES	Add device notes

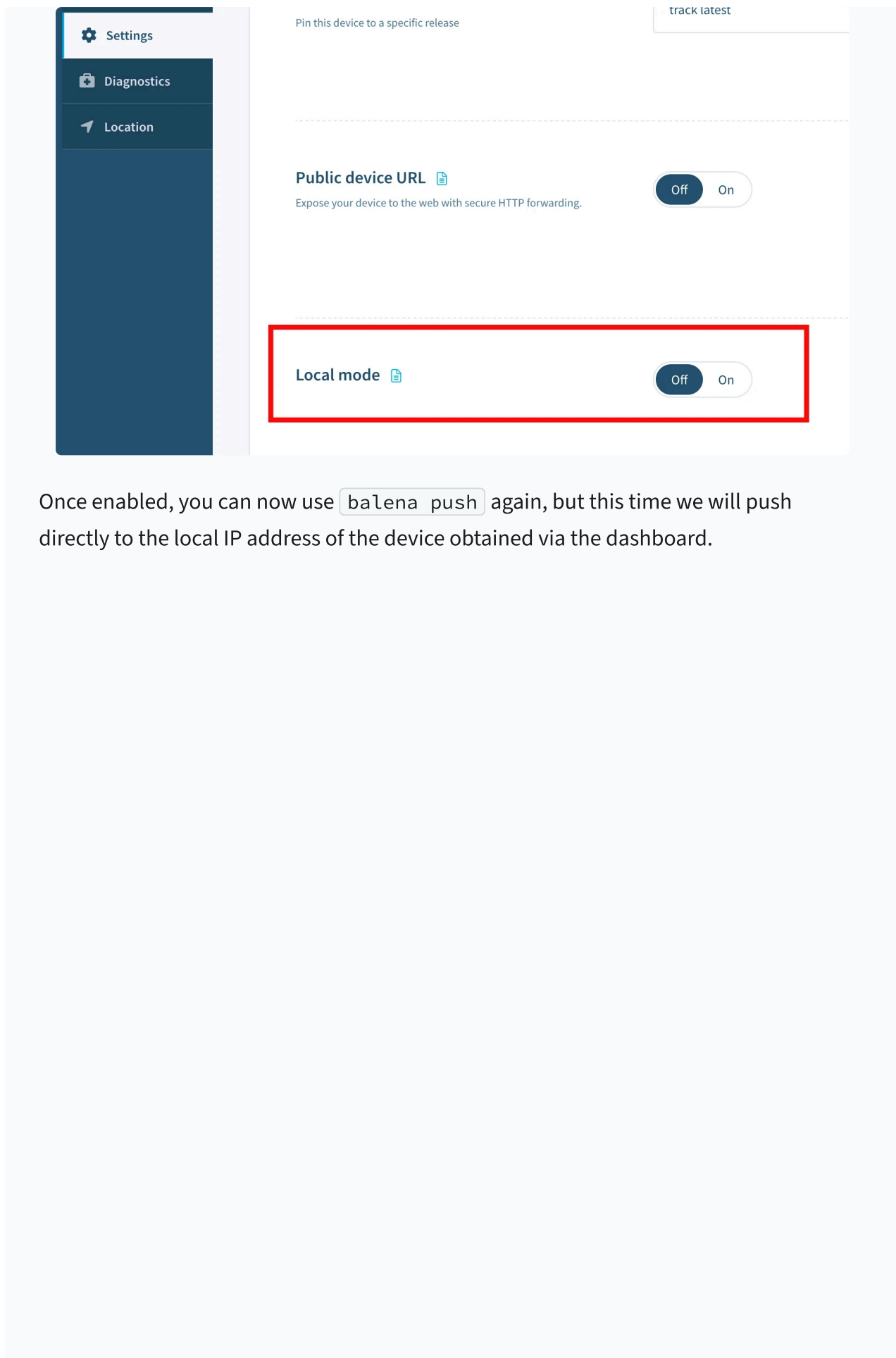
Follow the URL to view the welcome page with additional resources. Alternatively, you can point your browser to your device's local IP address to access the server running on your device. You can find the device's IP address on the device dashboard page. This is what you should be seeing.



Developing your project

Now, let's try making some changes to this project and testing them right on the device. The project can be modified and pushed again using the same method as above, but since we are using a development version of the OS, we can enable *Local mode* and push directly to the device for a faster development cycle.

Activate local mode on the device via the dashboard.



Once enabled, you can now use `balena push` again, but this time we will push directly to the local IP address of the device obtained via the dashboard.

first-device

Actions ▾

FLEET

unicorn_s_organization/first-fleet

STATUS	CLOUDLINK	CLOUDLINK LAST CONNECTED
Operational		Connected (for 1 minute)
HEARTBEAT	UUID	TYPE
	b36a040	
UPDATE STATUS	UPDATE STATUS DURATION	HOST OS VERSION
Done	1 minute	balenaOS 6.10.24+rev2 (Changelog ↗)
OS VARIANT	SUPERVISOR VERSION	CURRENT RELEASE
PRODUCTION	17.4.2	2.0.4+rev1
TARGET RELEASE	SUPPORT ACCESS	IS ACTIVE
2.0.4+rev1	Off	
LOCAL MODE	PUBLIC DEVICE URL	
LOCAL IP ADDRESS	PUBLIC IP ADDRESS	
192.168.1.187	47.20.198.80	
MAC ADDRESS		
B8:27:EB:C3:E9:03		
B8:27:EB:96:BC:56		
TAGS (0)		
No tags configured yet		
NOTES		
No notes		

```
$ balena push 10.19.0.153
```

The same build process as before is carried out, but this time instead of using the balena builders, the build takes place locally on the device itself.

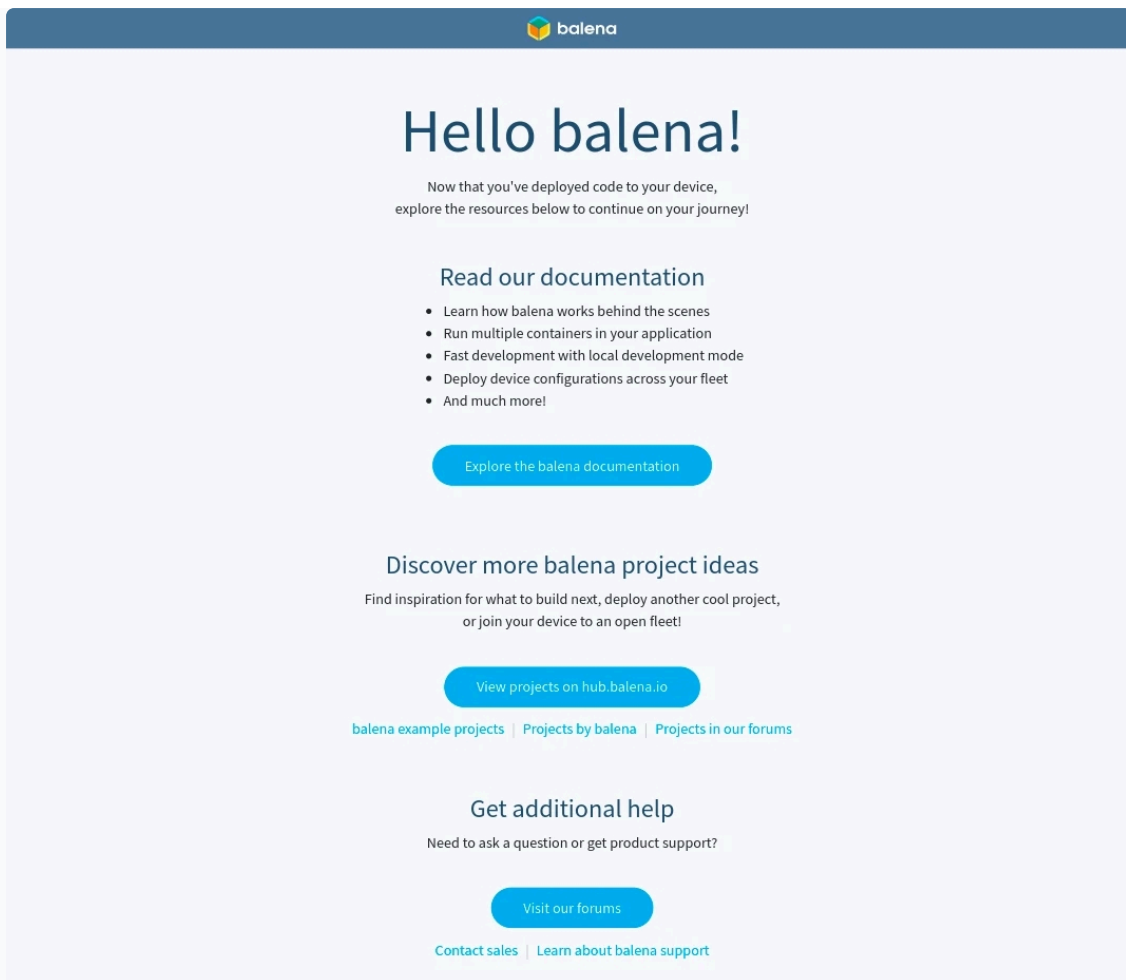
```
[Info]    Streaming device logs...
[Live]    Watching for file changes...
[Live]    Waiting for device state to settle...
[Logs]    [8/26/2021, 11:58:18 AM] Creating network 'default'
[Logs]    [8/26/2021, 11:58:19 AM] Installing service 'hello-
world sha256:...'
[Logs]    [8/26/2021, 11:58:20 AM] Installed service 'hello-world
sha256:...'
[Logs]    [8/26/2021, 11:58:20 AM] Starting service 'hello-world
sha256:...'
[Logs]    [8/26/2021, 11:58:23 AM] Started service 'hello-world
sha256:...'
[Logs]    [8/26/2021, 11:58:24 AM] [hello-world] Starting server
on port 80
[Live]    Device state settled
```

The balena CLI will now watch for changes to all the files within the project, and automatically push changes to the device when detected. Let's try making a change to title of our balena welcome page. Navigate to the `index.html` file present in the `static` directory of the project. Open the file and change the title from `Welcome to balena!` to `Hello balena!` and save the file. After saving the changes, you can observe balena CLI automatically start rebuilding only the parts of the Dockerfile that has been changed.

```
[Live]    Detected changes for container hello-world, updating...

...
[Live]    [hello-world] Restarting service...
[Logs]    [8/26/2021, 11:59:01 AM, 2:42:32 AM] Service exited
'hello-world sha256:...'
[Logs]    [8/26/2021, 11:59:04 AM, 2:42:35 AM] Restarting service
'hello-world sha256:...'
[Logs]    [8/26/2021, 11:59:05 AM, 2:42:36 AM] [hello-world]
Starting server on port 80
```

When the rebuild is complete, take a look at the public device URL again to see your changes. The welcome page should have been updated with the new title.



Next steps

Once you've finished making your changes, disable local mode and the device will revert back to running the latest release that's on your fleet. To update your fleet with the latest changes you've just worked on, use `balena push <fleet name>` once more to create a new release with those changes.

When it's finished building the device(s) will update as before. Remember anything pushed to the fleet in this way can be applied to 10+ or 1000+ devices with no extra effort! To continue learning, explore parts of the guide in more detail:

- Learn more about [local mode](#), which allows you to build and sync code to your device locally for rapid development.
- Develop an application with [multiple containers](#) to provide a more modular approach to fleet management.

- Manage your device fleet with the use of [configuration](#), [environment](#), and [service variables](#).
- Find out more about the [balena CLI](#) ↗ and the functionality it offers.
- Visit our blog to find step-by-step tutorials for some [classic balena projects](#) ↗.
- To publish what you will build or have already built, head over to [balenaHub](#) ↗.
- If you find yourself stuck or confused, help is just a [click away](#) ↗.

Enjoy balenafying all the things!

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Last updated 2 days ago

Was this helpful?

